

TOKE POINT, WA, USA

WMO: 994029

Lat: 46.7070N

Lon: 123.9670W

Elev: 5

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -0.8	(c) 0.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 16.9	(k) 12.1	(l) 14.9	(m) 10.0	(n) 5.6	(o) 70	(p) 0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.2	(c) 24.8	(d) N/A	(e) 22.3	(f) N/A	(g) 20.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.5	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.4	(b) 10.7	(c) 9.1	(e) -3.0	(f) 29.8	(g) 2.2	(h) 2.3	(i) -4.6	(j) 31.4	(k) -5.9	(l) 32.8	(m) -7.1	(n) 34.0	(o) -8.7	(p) 35.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	7.0	6.9	7.9	9.2	12.4	13.7	15.4	16.0	15.4	12.1	8.5	6.4
	DBStd	4.27	2.33	2.64	2.33	2.18	3.47	2.04	1.95	1.80	2.28	2.16	2.63	2.91
	HDD10.0	482	96	90	74	39	6	0	0	0	0	6	57	116
	HDD18.3	2725	352	322	324	274	191	141	94	76	94	192	294	371
	CDD10.0	822	2	2	8	14	81	112	167	187	161	72	13	3
	CDD18.3	22	0	0	0	0	8	2	3	4	4	0	0	0
	CDH23.3	136	0	0	0	1	49	16	25	21	23	1	0	0
	CDH26.7	32	0	0	0	0	18	3	7	2	2	0	0	0
Wind	WSAvg	4.2	5.0	4.6	4.7	4.1	3.6	3.7	3.5	3.0	3.6	4.2	5.0	5.1
Precipitation	PrecAvg	2157	349	215	245	158	90	66	25	44	74	215	329	345
	PrecMax	2741	695	612	510	294	158	132	70	133	232	433	671	598
	PrecMin	1547	173	24	44	63	13	15	5	3	7	31	94	153
	PrecStd	352	105	118	105	62	41	31	17	35	56	100	142	117
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	13.8	18.0	19.9	28.2	25.8	27.1	26.3	26.7	21.0	14.9	13.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	11.8	12.5	14.0	15.8	23.7	20.5	22.4	23.0	23.0	17.6	13.9	12.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	10.9	11.2	12.1	13.3	20.3	18.0	19.8	20.5	20.2	15.8	13.0	11.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	10.2	10.2	11.0	11.8	17.8	16.5	18.2	18.7	18.2	15.0	12.2	10.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.3	4.6	4.7	4.7	4.4	4.7	5.2	5.5	5.1	4.5	3.6
		MCDBR	5.3	5.8	6.8	8.2	9.3	8.5	8.9	9.7	10.1	6.8	4.8	3.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.331	0.321	0.338	0.370	0.399	0.373	0.353	0.357	0.347	0.339	0.331	0.322
	taud		2.471	2.543	2.508	2.385	2.310	2.418	2.507	2.494	2.526	2.540	2.495	2.483
	Ebn at Noon		759	858	891	886	867	891	906	889	870	824	753	726
	Edh at Noon		68	76	90	112	125	113	102	99	88	75	65	60
All-Sky Solar Radiation	RadAvg		1.07	1.94	2.94	4.27	5.24	5.72	6.03	5.20	4.03	2.37	1.28	0.90
	RadStd		0.15	0.24	0.33	0.38	0.45	0.48	0.48	0.42	0.34	0.26	0.16	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page